

CUTOFF FOR THE SHERRINGTON–KIRKPATRICK GLAUBER DYNAMICS AT $\beta < 1/3$

ABSTRACT. We prove quenched worst-case total-variation cutoff for rate-one-per-site heat-bath dynamics of the zero-field Sherrington–Kirkpatrick model for every fixed inverse temperature $0 < \beta < 1/3$. The transition window is bounded in this clock. The principal estimate propagates a dimension-independent variance-tensorization inequality along the evolving law from every product initialization. Its proof differentiates the optimal Poincaré constant of the heat-bath chain associated with the evolving law, preserves the signed interaction matrix in the principal term, and absorbs the remaining term into an exact negative square. The local-product argument then gives cutoff. We prove a logarithmic lower bound and recall the existing logarithmic upper order. The precise positive-temperature cutoff location is not identified.

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1. MODEL AND MAIN RESULT

For each integer $N \geq 2$, let J be symmetric, with zero diagonal and independent entries $J_{ij} \sim \mathcal{N}(0, 1/N)$ for $i < j$. Its law is denoted by $\mathbb{P}_J$. Fix $\beta > 0$ independently of N , and define the probability measure on $\Omega_N = \{-1, 1\}^N$

$$\mu_{N,\beta,J}(x) = \frac{1}{Z_{N,\beta,J}} \exp \left(\beta \sum_{i < j} J_{ij} x_i x_j \right). \quad (1.1)$$

Every site has an independent rate-one clock. When site i rings, its spin is resampled with mean

$$n_i(x) = \tanh \left(\beta \sum_{j \neq i} J_{ij} x_j \right).$$

Write L for the generator and P_t for its semigroup, with N, β, J suppressed when fixed. Thus time t corresponds to a Poisson number of single-site updates with mean Nt .

For probability measures on a finite set, put $\|\nu - \mu\|_{\text{TV}} = \frac{1}{2} \sum_y |\nu(y) - \mu(y)|$. Define

$$d_N^J(t) = \max_{x \in \Omega_N} \|P_t(x, \cdot) - \mu_{N,\beta,J}\|_{\text{TV}}, \quad t_{\text{mix}}^{N,J}(\varepsilon) = \inf\{t \geq 0 : d_N^J(t) \leq \varepsilon\}.$$

The maximum is taken after J is fixed.

Theorem 1.1. *For every fixed $0 < \beta < 1/3$ and $0 < \varepsilon < 1/2$,*

$$\frac{t_{\text{mix}}^{N,J}(\varepsilon)}{t_{\text{mix}}^{N,J}(1-\varepsilon)} \xrightarrow[N \rightarrow \infty]{\mathbb{P}_J} 1. \quad (1.2)$$

More precisely, put $a_\beta = 1 - 2\beta - 3\beta^2 > 0$. There are events $\mathcal{G}_{N,\beta}$, depending only on J , with $\mathbb{P}_J(\mathcal{G}_{N,\beta}) \rightarrow 1$, such that on these events, simultaneously for all $0 < \varepsilon < 1/2$,

$$0 \leq t_{\text{mix}}^{N,J}(\varepsilon) - t_{\text{mix}}^{N,J}(1-\varepsilon) \leq \frac{6}{a_\beta} \log \frac{1}{\varepsilon}, \quad (1.3)$$

$$t_{\text{mix}}^{N,J}(1-\varepsilon) \geq \frac{1}{2} \log N - \frac{1}{2} \log \frac{16}{a_\beta \varepsilon}. \quad (1.4)$$

For each fixed $\varepsilon \in (0, 1)$, the mixing time is also $O_{\beta,\varepsilon}(\log N)$ with high probability.

The proof covers disorder-dependent initial states. The bounded window does not identify a deterministic center accurate to order one, a leading constant in $c(\beta) \log N$, or a limiting profile at positive β . The theorem makes no assertion at $\beta = 1/3$ or throughout $\beta < 1/2$.

1.1. Proof strategy and inputs. For a full-support law ν , let $C(\nu)$ be the optimal constant in

$$\text{Var}_\nu f \leq C(\nu) \sum_i \nu[\text{Var}_\nu(f \mid x_{-i})].$$

This is the reciprocal of the rate-one spectral gap of the heat-bath chain with stationary law ν . In particular, a full-support product law has $C = 1$. We prove a deterministic bound on $C(\nu_0 P_t)$ for every product initial law, without assuming that the evolving law is an Ising measure.

To describe the bound, let $B = \beta J$, set

$$K = \|B\|_{\text{op}}, \quad M = \max_{i,j} |B_{ij}|, \quad R = \max_i \sum_j B_{ij}^2,$$

and define

$$k = K + (2e^{4M} + e^{8M})R. \quad (1.5)$$

If $k < 1$, Proposition 2.1 proves $C(\nu_0 P_t) \leq C_* = (1 - k)^{-1}$. Comparison of energies then gives a local Poincaré inequality for the original dynamics, with constant $(2C_*)^{-1}$.

The local-product method of Pedrotti and Salez [1] converts this into a window of width at most $3C_* \log(1/\varepsilon)$. For transparency, Section 3 includes the finite-state chi-square calculation and overlap argument. A linear observable gives $d_N^J(t) \geq 1 - 8C_* e^{2t}/N$. Finally, the SK matrix estimates yield $k \leq 2\beta + 3\beta^2 + o(1)$ with high probability.

The matrix estimates and the additional logarithmic upper order are verified in Wang [2], Lemma 3.5 and Theorem 1.1, respectively. The latter is an order-of-magnitude input; it does not supply the cutoff window. All remaining arguments below are deterministic.

2. A POINCARÉ INEQUALITY FOR THE EVOLVING LAW

2.1. The two heat-bath operators. Fix a symmetric zero-diagonal matrix B and a vector $b \in \mathbb{R}^N$. This section permits arbitrary fixed external fields. The target measure is

$$\mu_{B,b}(x) \propto \exp\left(\frac{1}{2}x^\top Bx + b \cdot x\right).$$

Its conditional means are $n_i(x) = \tanh(b_i + \sum_j B_{ij}x_j)$, which are independent of x_i . All update probabilities are strictly positive.

Let x^i denote x with spin i flipped, and define

$$\partial_i f(x) = \frac{f(x) - f(x^i)}{2x_i}.$$

This function is independent of x_i . The target generator and its carré du champ are

$$Lf = \sum_i (n_i - x_i) \partial_i f, \quad (2.1)$$

$$\Gamma f = \frac{1}{2}L(f^2) - fLf = \sum_i (1 - x_i n_i) (\partial_i f)^2. \quad (2.2)$$

Given a full-support probability measure ν on Ω_N , define its conditional expectation operators and conditional means by

$$Q_i f = \nu[f \mid x_{-i}], \quad D_i f = f - Q_i f, \quad m_i = \nu[x_i \mid x_{-i}], \quad v_i = 1 - m_i^2.$$

These objects refer to ν , whereas n_i and L refer to the fixed target. We have $D_i f = (x_i - m_i) \partial_i f$ and

$$\mathcal{E}_\nu(f) := \sum_i \nu[(D_i f)^2] = \sum_i \nu[v_i (\partial_i f)^2]. \quad (2.3)$$

The operator $\mathcal{A}_\nu = \sum_i (I - Q_i)$ is positive selfadjoint in $L^2(\nu)$; its nullspace consists of constants. Define

$$C(\nu) = \sup_{f \text{ nonconstant}} \frac{\text{Var}_\nu f}{\mathcal{E}_\nu(f)}. \quad (2.4)$$

The quotient is invariant under addition of constants. Every centered extremizer satisfies

$$\sum_i D_i f = C(\nu)^{-1} f. \quad (2.5)$$

Proposition 2.1. *Let K, M, R, k be as in (1.5), and suppose $k < 1$. For every product initial law ν_0 , including a point mass,*

$$\text{Var}_{\nu_0 P_t} f \leq \frac{1}{1-k} \mathcal{E}_{\nu_0 P_t}(f) \quad (t \geq 0). \quad (2.6)$$

For a law without full support, the energy is interpreted by continuous extension, as specified at the end of the proof. In particular, $C(\nu_0 P_t) \leq (1-k)^{-1}$ for every $t > 0$.

2.2. Exact differentiation of the variational quotient. Fix a time at which $\nu = \nu_t = \nu_0 P_t$ has full support, and put $C = C(\nu)$. Choose a centered extremizer f satisfying (2.5). In all derivatives in the next lemma, f is held fixed at this time. Let

$$V = \text{Var}_\nu f, \quad E = \mathcal{E}_\nu(f), \quad A = \nu[\Gamma f].$$

Define the nonnegative square sum T and the interaction term S by

$$T = \sum_{i \neq j} \nu[(1 - x_j n_j)(\partial_j D_i f)^2], \quad (2.7)$$

$$S = \sum_{i \neq j} \nu[v_i \partial_i f (\partial_i n_j) \partial_j Q_i f]. \quad (2.8)$$

Lemma 2.2. *At the fixed time above,*

$$V' - CE' = -2(C-1)A - 2CT + 2CS. \quad (2.9)$$

Proof. The master equation and centering of f give $V' = \nu[L(f^2)]$. On differentiating the own energy, the derivatives of the conditional means vanish by orthogonality:

$$E' = \sum_i \nu[L((D_i f)^2)] - 2 \sum_i \nu[D_i f \dot{Q}_i f] = \sum_i \nu[L((D_i f)^2)].$$

Indeed, $\dot{Q}_i f$ is independent of x_i and $\nu[D_i f \mid x_{-i}] = 0$.

Expand $L(g^2) = 2gLg + 2\Gamma g$, use $LD_i f = Lf - LQ_i f$, and sum (2.5). The terms involving fLg cancel, leaving

$$\begin{aligned} V' - CE' &= 2\nu[\Gamma f] - 2C \sum_i \nu[\Gamma(D_i f)] \\ &\quad + 2C \sum_i \nu[D_i f LQ_i f]. \end{aligned} \quad (2.10)$$

Since $\partial_i D_i f = \partial_i f$, the middle sum is $A + T$. The $j = i$ summand in $LQ_i f$ is zero. For $j \neq i$, the functions $\partial_i f$ and $\partial_j Q_i f$ are independent of x_i . Write the function n_j , on this two-point fiber, as its average plus $x_i \partial_i n_j$. Then

$$\nu[(x_i - m_i)(n_j - x_j) \mid x_{-i}] = (1 - m_i^2) \partial_i n_j.$$

Using $D_i f = (x_i - m_i) \partial_i f$ shows that the last sum in (2.10) is S . This proves the identity. ■

This calculation uses conditional expectation under the actual law ν_t . It neither freezes that law as an Ising measure nor conditions Gaussian disorder variables.

2.3. Global bounds for the target derivative. Put

$$s_j = 1 - n_j^2 = \operatorname{sech}^2 \left(b_j + \sum_{\ell} B_{j\ell} x_{\ell} \right).$$

Lemma 2.3. *For all configurations and all $i \neq j$,*

$$\partial_i n_j = B_{ij} s_j + \rho_{ij}, \quad |\rho_{ij}| \leq 2e^{4M} B_{ij}^2 s_j, \quad (2.11)$$

$$|\partial_i n_j| \leq e^{4M} |B_{ij}| s_j. \quad (2.12)$$

Proof. Let $u = b_j + \sum_{\ell} B_{j\ell} x_{\ell}$. Symmetry of B and the exact field change give

$$\partial_i n_j = B_{ij} \int_0^1 \operatorname{sech}^2(u - 2r B_{ij} x_i) dr.$$

For $g(u) = \operatorname{sech}^2 u$, one has $|(\log g)'| \leq 2$ and $|g'| \leq 2g$. Consequently g along the displayed segment is at most $e^{4M} s_j$, which proves (2.12). Moreover,

$$|g(u - 2r B_{ij} x_i) - g(u)| \leq 4r |B_{ij}| e^{4M} s_j.$$

Integrating over r and multiplying by $|B_{ij}|$ proves (2.11). ■

These are relative bounds valid for every real u . There is no restriction on the magnitude of a local field or of b .

2.4. The signed bound and square completion. For the remainder of this proof put $a_i = \partial_i f$ and $w_i = 1 - m_i n_i$. Conditioning on x_{-i} gives

$$A = \nu \left[\sum_i w_i a_i^2 \right]. \quad (2.13)$$

The following elementary inequalities are used pointwise:

$$v_i^2 + s_i^2 \leq v_i + s_i = 2 - m_i^2 - n_i^2 \leq 2w_i, \quad (2.14)$$

$$\frac{s_j^2}{1 - x_j n_j} = s_j(1 + x_j n_j) \leq 2. \quad (2.15)$$

The denominator is positive because $|n_j| < 1$.

Use $\partial_j Q_i f = a_j - \partial_j D_i f$ to write $S = S_0 - S_1$. The principal contribution to S_0 in (2.11) is $u^T B z$, where $u_i = v_i a_i$ and $z_i = s_i a_i$. For every configuration,

$$|u^T B z| \leq \frac{K}{2} (\|u\|_2^2 + \|z\|_2^2) \leq K \sum_i w_i a_i^2. \quad (2.16)$$

The entrywise square $B \circ B = (B_{ij}^2)$ is nonnegative and symmetric, with operator norm at most its maximum row sum R . Apply the same norm estimate to $|u|^T (B \circ B) |z|$. The remainder estimate in (2.11) then gives

$$|S_0| \leq (K + 2e^{4M} R) A. \quad (2.17)$$

In particular, the principal bound uses $\|B\|_{\text{op}}$, not the norm of the entrywise absolute matrix.

Weighted Cauchy–Schwarz gives $|S_1|^2 \leq T U$, where

$$U = \sum_{i \neq j} \nu \left[\frac{(\partial_i n_j)^2 v_i^2 a_i^2}{1 - x_j n_j} \right].$$

By (2.12), (2.15), the row-square bound, and (2.14),

$$U \leq 2e^{8M} R \nu \left[\sum_i v_i^2 a_i^2 \right] \leq 4e^{8M} R A.$$

It follows that

$$|S_1| \leq 2e^{4M} \sqrt{RAT}. \quad (2.18)$$

Substitute (2.17) and (2.18) into (2.9):

$$\begin{aligned} V' - CE' &\leq -2(C-1)A + 2C(K + 2e^{4M}R)A - 2CT + 4Ce^{4M}\sqrt{RAT} \\ &\leq -2[(1-k)C-1]A. \end{aligned} \quad (2.19)$$

For the last step use $-T + 2e^{4M}\sqrt{RAT} \leq e^{8M}RA$.

2.5. The invariant bound. We give the finite-dimensional justification for differentiating (2.4). Fix one state x_0 and restrict the quotient to the fixed vector space $\{f : f(x_0) = 0\}$. On its Euclidean unit sphere, the denominator is positive for every full-support law. On a compact time interval along a full-support evolution, it is uniformly bounded away from zero. The quotient is therefore uniformly smooth in time on this compact sphere, and its maximum $C(t) = C(\nu_t)$ is locally Lipschitz.

The upper right derivative of this maximum is at most the largest time derivative of the quotient among its current maximizers. To see this, choose maximizers at times $t+h$, take a convergent subsequence on the sphere, and use uniform differentiability of the quotient. The limit is a maximizer at t . Subtracting its ν_t -mean does not change the quotient or its time derivative. Thus (2.19), which holds for every centered extremizer, gives the required bound even when the smallest positive eigenvalue has multiplicity.

The inequality $v_i \leq 2w_i$ follows from $2w_i - v_i = (m_i - n_i)^2 + 1 - n_i^2 \geq 0$. Hence $A \geq E/2$. Whenever $C(t) \geq C_* = (1-k)^{-1}$, (2.19) implies, at almost every such time,

$$C'(t) \leq -(1-k)(C(t) - C_*). \quad (2.20)$$

The positive part $(C(t) - C_*)_+$ is locally absolutely continuous. Its derivative satisfies the same linear upper bound almost everywhere, so Gronwall's inequality prevents it from becoming positive if it starts at zero.

For a full-support product initial law, the product variance inequality gives $C(\nu_0) \leq 1$, and a nonconstant single-coordinate function gives equality. Since $C_* \geq 1$, this proves (2.6) for such initial laws.

For a possibly singular product law, approximate each marginal by strictly positive marginals. The evolved laws converge at every fixed time, and their variances converge. On an unordered coordinate edge with endpoint probabilities p, q , the contribution to $\mathcal{E}_\nu(f)$ is

$$\frac{pq}{p+q} [f(x) - f(x^i)]^2,$$

with value zero when $p+q=0$. This is a continuous function of $(p, q) \in [0, 1]^2$. The finite sum defining the energy therefore converges too. Passing to the limit proves the claimed inequality, including at time zero. For $t > 0$, irreducibility makes the evolved law full support. This completes the proof of Proposition 2.1.

3. FROM LOCAL POINCARÉ TO A MIXING WINDOW

The comparison $v_i \leq 2w_i$ used above proves, for any full-support ν , $\mathcal{E}_\nu(f) \leq 2\nu[\Gamma f]$. Consequently Proposition 2.1 yields, uniformly in x, t, f ,

$$\text{Var}_{P_t(x, \cdot)} f \leq 2C_* P_t \Gamma f(x). \quad (3.1)$$

At $t = 0$ the variance is zero. Letting $t \rightarrow \infty$ also gives the same inequality under the stationary law $\mu_{B,b}$.

We recall the finite-state argument behind the local-product theorem of Pedrotti and Salez [1, Theorem 1 and Lemma 1]. For laws $\eta \ll \nu$, write $\chi^2(\eta\|\nu) = \sum_y \eta(y)^2/\nu(y) - 1$, with $0/0 = 0$.

Lemma 3.1. *Let $\gamma > 0$ and $\eta \ll \nu$. Suppose $\nu_u = \nu P_u$ satisfies $\text{Var}_{\nu_u} g \leq \gamma^{-1} \nu_u[\Gamma g]$ for all $u \geq 0$. Then*

$$\chi^2(\eta P_s\|\nu P_s) \leq e^{-2\gamma s} \chi^2(\eta\|\nu).$$

Proof. For positive reference laws let $r_u = (\eta P_u)/(\nu P_u)$, pointwise on the state space. If $q(x, y)$ are the generator entries, differentiating the two evolving laws and using $\sum_y q(x, y) = 0$ gives

$$\frac{d}{du} \sum_y \frac{(\eta P_u)(y)^2}{(\nu P_u)(y)} = - \sum_{x,y} \nu_u(x) q(x, y) (r_u(y) - r_u(x))^2 = -2\nu_u[\Gamma r_u].$$

Since $\nu_u r_u = 1$, the assumed inequality bounds the last expression by $-2\gamma \chi^2(\eta P_u\|\nu_u)$. Integration proves the claim. If the initial reference law has zeros, apply the argument from a positive time δ . On the finite state space, $\eta \ll \nu$ implies $\eta \leq M\nu$ for some finite M , and hence $\eta P_\delta \leq M\nu P_\delta$. If $\nu(y) = 0$, then

$$\frac{(\eta P_\delta)(y)^2}{(\nu P_\delta)(y)} \leq M^2 (\nu P_\delta)(y) \rightarrow 0.$$

The other summands converge ordinarily. Thus the initial chi-square divergence is continuous as $\delta \downarrow 0$, proving the claim. $\blacksquare$

Fix an initial state x and a time t , and set $\nu = P_t(x, \cdot)$ and $\mu = \mu_{B,b}$. If $\|\nu - \mu\|_{\text{TV}} \leq 1 - \varepsilon$, the common part $\min(\nu, \mu)$ has mass $a \geq \varepsilon$. Its normalization η satisfies $\eta/\nu \leq 1/a$ and $\eta/\mu \leq 1/a$. Thus $\chi^2(\eta\|\nu), \chi^2(\eta\|\mu) \leq 1/a - 1 \leq 1/\varepsilon$. Apply Lemma 3.1 twice, using (3.1) along the future of ν and under μ , with $\gamma = (2C_*)^{-1}$. The inequality $2\|\rho - \lambda\|_{\text{TV}} \leq \sqrt{\chi^2(\rho\|\lambda)}$ and the triangle inequality give

$$\|P_{t+s}(x, \cdot) - \mu\|_{\text{TV}} \leq \varepsilon^{-1/2} e^{-s/(2C_*)}. \quad (3.2)$$

If $t = t_{\text{mix}}(1 - \varepsilon)$ for the worst-case distance, the hypothesis holds for every x . Taking the maximum in (3.2) and then setting $s = 3C_* \log(1/\varepsilon)$ proves

$$t_{\text{mix}}(\varepsilon) - t_{\text{mix}}(1 - \varepsilon) \leq 3C_* \log \frac{1}{\varepsilon}. \quad (3.3)$$

All functions on the finite cube belong to the generator domain. The chain is irreducible and converges to its stationary measure. Thus this is also a direct verification of the hypotheses and clock normalization of the cited local-product theorem.

4. A LOWER BOUND FROM THE SPIN OBSERVABLES

In this section set $b = 0$. Spin-flip symmetry gives $\mu x_i = 0$. For any vector $u \in \mathbb{R}^N$, the oriented derivatives of $u \cdot x$ are the constants u_i . Since $v_i \leq 1$, Proposition 2.1 and its stationary limit yield

$$\text{Var}_{P_t(x, \cdot)}(u \cdot \sigma) \leq C_* \|u\|_2^2, \quad \text{Var}_\mu(u \cdot \sigma) \leq C_* \|u\|_2^2. \quad (4.1)$$

The original chain is reversible with respect to μ . Let $\mathcal{A} = -L$. The spectral measure of the function σ_i is a probability measure, because $\mu[\sigma_i^2] = 1$. Its first moment is

$$\langle \sigma_i, \mathcal{A} \sigma_i \rangle_\mu = \mu[1 - n_i^2] \leq 1.$$

Jensen's inequality for $z \mapsto e^{-2tz}$ gives

$$\|P_t \sigma_i\|_{L^2(\mu)}^2 = \langle \sigma_i, e^{-2t\mathcal{A}} \sigma_i \rangle_\mu \geq e^{-2t}.$$

Summing over i and choosing a state with value at least its μ -average, we obtain an $x = x(B, t)$ with mean vector

$$u_i = P_t \sigma_i(x), \quad \|u\|_2^2 \geq N e^{-2t}.$$

The observable $F(\sigma) = u \cdot \sigma$ has mean $\|u\|_2^2$ under $P_t(x, \cdot)$ and zero under μ . Chebyshev's inequality applied to $F > \|u\|_2^2/2$, using (4.1), proves

$$d_N^J(t) \geq 1 - \frac{8C_*}{\|u\|_2^2} \geq 1 - \frac{8C_* e^{2t}}{N}. \quad (4.2)$$

Consequently

$$t_{\text{mix}}(1 - \varepsilon) \geq \frac{1}{2} \log N - \frac{1}{2} \log \frac{8C_*}{\varepsilon}. \quad (4.3)$$

The chosen state and observable may depend on the disorder and on time. This is allowed by the worst-case definition.

5. THE SK EVENT AND PROOF OF CUTOFF

For the normalization in (1.1), the following estimates hold with probability tending to one:

$$\|J\|_{\text{op}} \leq 2 + o(1), \quad \max_i \sum_j J_{ij}^2 \leq 1 + o(1), \quad \max_{i,j} |J_{ij}| = o(1). \quad (5.1)$$

Here the error terms may be chosen as deterministic sequences tending to zero on a common event. The estimates, with the factor β absorbed into the interaction matrix, are verified in [2, Lemma 3.5 and its proof]. The entry estimate follows from a Gaussian union bound; the row estimate follows from chi-square concentration and a union bound. For the norm estimate, the GOE upper edge is $2 + o(1)$, and deleting its diagonal changes the norm by at most its maximum absolute diagonal entry, which is $o(1)$.

With $B = \beta J$, formula (1.5) becomes

$$\begin{aligned} k_N &= \beta \|J\|_{\text{op}} \\ &+ \left(2e^{4\beta \max |J_{ij}|} + e^{8\beta \max |J_{ij}|} \right) \beta^2 \max_i \sum_j J_{ij}^2 \leq 2\beta + 3\beta^2 + o(1). \end{aligned} \quad (5.2)$$

Fix $0 < \beta < 1/3$. Since $a_\beta = 1 - 2\beta - 3\beta^2 > 0$, we may take $\mathcal{G}_{N,\beta}$ to be the event on which

$$1 - k_N \geq a_\beta/2, \quad C_* = (1 - k_N)^{-1} \leq 2/a_\beta.$$

Its probability tends to one by (5.2). On this single event the deterministic proposition holds for all initial states and all times; no union bound over the 2^N states is used. Substitution

into (3.3) and (4.3) proves (1.3) and (1.4). For each fixed $\varepsilon \in (0, 1/2)$ and all sufficiently large N ,

$$0 \leq \frac{t_{\text{mix}}(\varepsilon)}{t_{\text{mix}}(1-\varepsilon)} - 1 \leq \frac{(6/a_\beta) \log(1/\varepsilon)}{\frac{1}{2} \log N - \frac{1}{2} \log(16/(a_\beta \varepsilon))} \rightarrow 0.$$

This proves the cutoff assertion of Theorem 1.1.

5.1. The logarithmic upper order and the clock. Wang [2, Theorem 1.1] proves that for every fixed $\beta < 1/2$, with high probability over the disorder, random-scan discrete heat-bath dynamics mixes from every state in $O_\beta(N \log(N/\varepsilon))$ updates, uniformly over external fields. Let K_d denote that discrete kernel. Our clock gives

$$P_t = e^{Nt(K_d - I)} = \mathbb{E}[K_d^{Z_t}], \quad Z_t \sim \text{Poisson}(Nt).$$

Choose an integer $r = O_\beta(N \log(N/\varepsilon))$ with discrete distance at most $\varepsilon/2$. Convexity of total variation, monotonicity of the discrete distance, and a Poisson lower-tail bound give

$$d_N^J(2r/N) \leq \varepsilon/2 + \mathbb{P}(\text{Poisson}(2r) < r) \leq \varepsilon$$

for all sufficiently large N at fixed ε . This proves the remaining upper-order statement in Theorem 1.1. It is separate from the window proof.

6. SCOPE OF THE RESULT

The main deterministic estimate is Proposition 2.1. It allows arbitrary external fields, but zero field is used in the spin-observable lower bound for the SK theorem. No monotone coupling, censoring inequality, Gaussian independence after a disorder-dependent choice, small-local-field assumption, or pointwise Bakry–Émery curvature estimate is used.

The criterion displayed here has limiting bound $2\beta + 3\beta^2$, which reaches one at $\beta = 1/3$. Extending this proof throughout $\beta < 1/2$ would require a sharper estimate than (2.17)–(2.19), or a different propagation argument. Neither that extension nor the precise positive-temperature location is a conclusion of this manuscript.

APPENDIX A. THE INDEPENDENT-SPIN CLOCK CHECK

This appendix sets $\beta = 0$, so the heat-bath coordinates evolve independently. Starting from all plus, the coordinates are independent and each has mean e^{-t} . Symmetry makes the distance the same from every initial state.

Fix $s \in \mathbb{R}$, put $t = \frac{1}{2} \log N + s$ and $a = e^{-s}$. Relative to the uniform measure, the density is

$$R_N(\sigma) = \prod_{i=1}^N (1 + a\sigma_i/\sqrt{N}).$$

Taylor expansion and the central limit theorem show $\log R_N \Rightarrow aZ - a^2/2$ with $Z \sim \mathcal{N}(0, 1)$. Furthermore $\mathbb{E}_{\text{unif}} R_N^2 = (1 + a^2/N)^N \leq e^{a^2}$. Uniform integrability therefore justifies passage to the limit in $\frac{1}{2} \mathbb{E}_{\text{unif}} |R_N - 1|$, yielding

$$d_N \left(\frac{1}{2} \log N + s \right) \rightarrow \mathbb{E}[(e^{aZ - a^2/2} - 1)_+] = 2\Phi(a/2) - 1,$$

where Φ is the standard normal distribution function. The center is $\frac{1}{2} \log N$ and the window has order one.

For the discrete chain choosing one site uniformly at each step, the Walsh character indexed by a set of size k has eigenvalue $1 - k/N$. Expanding the point-mass density in this orthonormal basis, at any positive integer step r ,

$$4d_{\text{d}}(r)^2 \leq \sum_{k=1}^N \binom{N}{k} (1 - k/N)^{2r} \leq (1 + e^{-2r/N})^N - 1.$$

Starting from all plus, the total magnetization has mean $N(1 - 1/N)^r$ and variance at most N : for two distinct coordinates, their product has mean $(1 - 2/N)^r \leq (1 - 1/N)^{2r}$. Its equilibrium variance is N as well. The last display gives the upper bound at $r = (N/2) \log N + cN$, and Chebyshev's inequality gives the lower bound at $r = (N/2) \log N - cN$ as $c \rightarrow \infty$. Thus the discrete center is $(N/2) \log N$, with window of order N . The Poisson count fluctuation at this scale is $O_{\mathbb{P}}(\sqrt{N \log N}) = o_{\mathbb{P}}(N)$. The positive-temperature theorem above is stated in continuous time; the only general-model conversion used is the one-sided Poissonization bound in Section 5.

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